

Assignment 1

Shravanya Andhale

D20A

Explore following 4 websites and answer the questions

- bitcoin.org
- [Blockchain.com](https://blockchain.com/)
- <https://coinmarketcap.com/>
- <https://www.investopedia.com/>

One-Word Answerable Questions from bitcoin.org

Bitcoin Basics

1. Bitcoin was introduced by whom? **Satoshi Nakamoto**
2. Bitcoin transactions are recorded on which data structure? **Blockchain**
3. Bitcoin uses which consensus mechanism? **Proof of Work Consensus**
4. The smallest unit of Bitcoin is called? **Satoshi**
5. Bitcoin is a **peer-to-peer** type of what system? **Electronic Cash System**

Wallets

6. A wallet stores the user's **private** key.
7. The recovery phrase used in wallets is called a **seed/mnemonic** phrase.
8. Hardware wallets store keys **offline**(online/offline).

9. Bitcoin addresses are derived from **public** keys.
10. What is required to sign a transaction? **private** (public/private)

Security

11. Bitcoin prevents double-spending using **Proof of Work Consensus mechanism and public ledger.**
12. Bitcoin is secured using **asymmetric** cryptography (asymmetric/symmetric).
13. To run a Bitcoin node, the software required is Bitcoin **Core**.
14. The process of verifying transactions is called **Proof of Work Consensus**.
15. The Bitcoin network is described as **decentralized** (centralized/decentralized).

Mining

16. Miners solve **cryptographic** puzzles to add blocks.
17. Mining rewards a block with newly created **Bitcoins**.
18. The block time target for Bitcoin is **10** minutes.
19. Mining difficulty adjusts every **2,016** blocks.
20. Miners compete to find a valid block **hash** (hash/signature).

Blockchain Structure

21. Each block contains a **Merkle** root.
22. Blocks are linked using the previous block's **hash**.
23. A Bitcoin transaction contains inputs and **outputs**.
24. Unspent transaction outputs are called **UTXO**.
25. The first-ever Bitcoin block is called the **Genesis** block.

Network

26. A full node stores the entire blockchain history/ledger.
27. Bitcoin uses which protocol for communication? P2P (TCP/P2P)
28. A light wallet uses SPV mode (SPV/full).
29. Nodes relay information across a peer-to-peer network.
30. The maximum Bitcoin supply is 21 million.

Task B1: Explore a Block

Note the following from the **latest block**:

- Block height: **960833**
- Block hash: **00000000000000000000000031f27be35daba44eecd10b82776d82f77b8e80e976c7**
- Timestamp: **03 Aug 2026, 12:01:30**
- Number of transactions: **5,516**
- Miner/pool name: **Unknown**

Task B2: Explore a Transaction

Open any one Bitcoin transaction.

10. Write the TXID of the transaction.
b50daac2007959ddb7805734c40544c56aa4ba7e1a8bcf0583b920eece0a7ff3
11. Number of inputs: **2**
12. Number of outputs: **1**
13. Transaction fee: **169SATS• \$0.11**
14. How many confirmations does it have? **960,835**

Task B3: Explore an Address

15. Search for any Bitcoin address shown in the block.

bc1q5nr4jv3vf0pf8dwgempqvqc7pl2zht5ywr6ln8

16. Record:

- Total BTC received: **52.32271508 BTC \$3,274,308**
- Total BTC sent: **52.32271508 BTC \$3,274,308**
- Current balance: **52.32271508 BTC \$3,274,308**

What is the role of a full node? **enforce the consensus rules of the blockchain by independently validating every single block and transaction**

List any two developer resources available on bitcoin.org.

Bitcoiner.Guide

Satoshi Nakamoto Institute

Explore : <https://coinmarketcap.com/>

Task C1: Market Information

- Current price of BTC: **\$62,555.11**
- 24-hour % change: **1.37%**
- Market capitalization: **\$1.25T**
- Circulating supply: **20.06M BTC**
- Total/maximum supply: **Total supply 20.06M BTC Max. supply 21M BTC**
- **Task C2: Historical Data**
- What was the price of Bitcoin one year ago? **90,424.58 dollar on 12/9/2025**

- Note the highest and lowest prices in the last 30 days. Highest- 66855 dollars lowest 61630 dollars

Explore <https://www.investopedia.com/>

Why is Bitcoin considered volatile?

ans:

Bitcoin is considered highly volatile because it is a relatively young asset driven by limited market liquidity, intense speculation, concentrated ownership among "whales," and hair-trigger investor sentiment. Because the market lacks the deep, institutional liquidity of traditional stock markets, large trades can trigger dramatic, sudden shifts in price

List two investment risks mentioned.

ans:

1. **Regulatory Risk:** The laws governing cryptocurrency continue to evolve globally. Changes in government policies, taxation rules, or outright trading bans could severely disrupt adoption and crush asset values.
2. **Limited Investor Protections (Security/Counterparty Risk):** If you store your crypto on an exchange, you face counterparty risk if that platform suffers a hack, operational failure, or bankruptcy. If you hold your own keys and lose them, or get hacked, there is generally no legal recourse or insurance to recover the stolen assets.

♦ Part E: Reflection & Analysis

Which website gave you the most clear understanding? Why?

Ans:

Blockchain.com and **Learnmeabitcoin.com** gave the best understanding as every new term was linked to its own page, examples were clear and statistics with visuals helped me understand.

In one paragraph, explain how blockchain ensures trust without a central authority.

Ans:

Blockchain ensures trust across a network by replacing centralized middlemen with a decentralized peer-to-peer network and automated math. Every node on the network maintains an identical copy of the ledger, and new transactions must be validated by network consensus using strict cryptographic rules. Once a block of transactions is confirmed, it is securely locked and chained to the previous block using unchangeable cryptographic hashes. This structure makes altering past records mathematically impossible ensuring transparency.